

Lecture 1: Introduction to electric components used in hybrid and electric vehicles

Electric Vehicle (EV) Configurations

Compared to HEV, the configuration of EV is flexible. The reasons for this flexibility are:

- The energy flow in EV is mainly via flexible electrical wires rather than bolted flanges or rigid shafts. Hence, distributed subsystems in the EV are really achievable.
- The EVs allow different propulsion arrangements such as independent four wheel and in wheel drives.

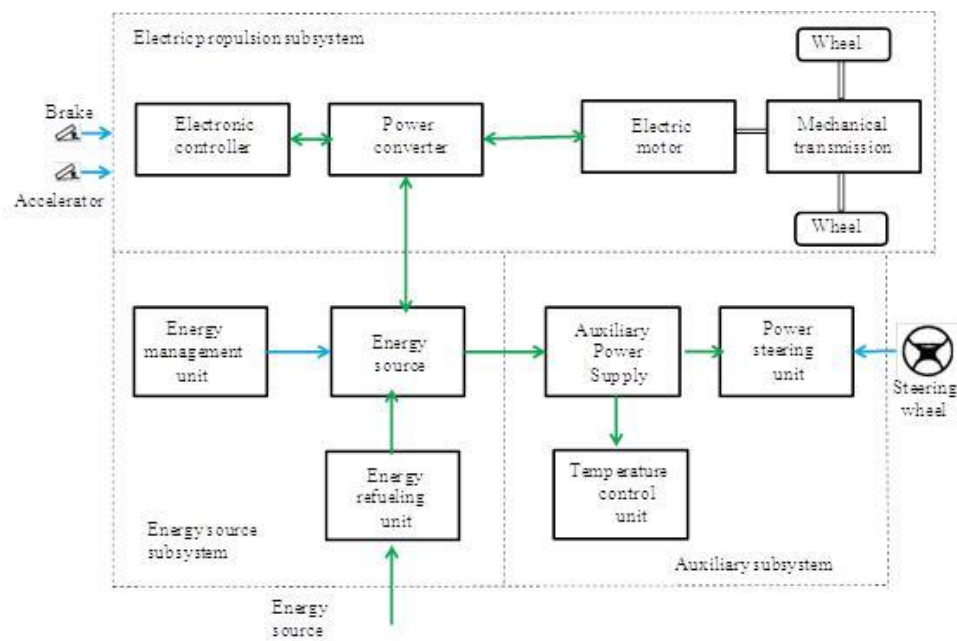


Figure 1: General Configuration of the EV

The EV has three major subsystems:

- Electric propulsion
- Energy source
- Auxiliary system

The electric propulsion subsystem comprises of: ➤ The electronic controller ➤ Power converter ➤ Electric Motor (EM) ➤ Mechanical transmission ➤ Driving wheels	The energy source subsystem consists of ➤ The energy source (battery, fuel cell, ultracapacitor) ➤ Energy management unit ➤ Energy refueling unit	The auxiliary subsystem consists of ➤ Power steering unit ➤ Temperature control unit ➤ Auxiliary power supply
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In **Figure 1** the black line represents the mechanical link, the green line represents the electrical link and the blue line represents the control information communication. Based on the control inputs from the brake and accelerator pedals, the electronic controller provides proper control signals to switch on or off the power converter which in turn regulates the power flow between the electric motor and the energy source. The backward power flow is due to regenerative braking of the EV and this regenerative energy can be stored provided the energy source is receptive.

The energy management unit cooperates with the electronic controller to control regenerative braking and its energy recovery. It also works with the energy-refueling unit to control refueling and to monitor usability of the energy source.

The auxiliary power supply provides the necessary power with different voltage levels for all EV auxiliaries, especially the temperature control and power steering units.

Functional block diagram of a typical electric propulsion system:

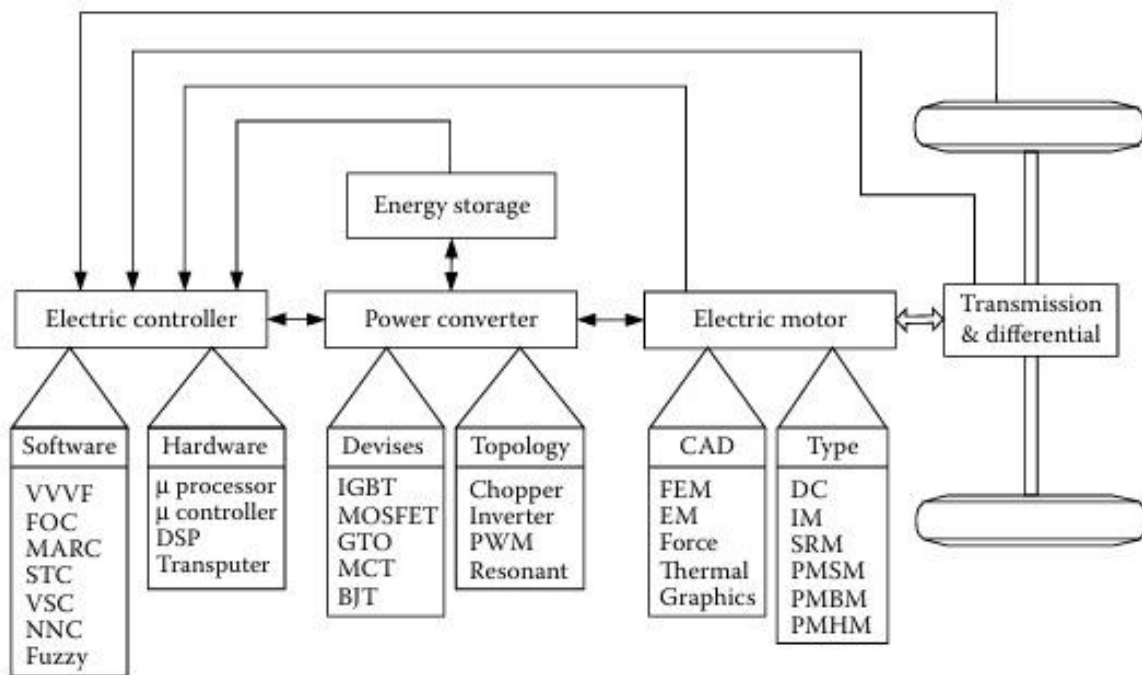


Figure 2: Functional block diagram of a typical electric propulsion system.

Electric propulsion systems are at the heart of EVs and HEVs.

Electric Motors	The electric motor converts the electric energy into mechanical energy to propel the vehicle or vice versa, to enable regenerative braking and/or to generate electricity for the purpose of charging the on-board energy storage.
Power Converters	The power converter is used to supply the electric motor with proper voltage and current.
Electronics Controllers	The electronic controller commands the power converter by providing control signals to it, and then controls the operation of the electric motor to produce proper torque and speed, according to the command from the driver.

The electronic controller can be further divided into three functional units

Sensor	The sensor is used to translate the measurable quantities, such as current, voltage, temperature, speed, torque, and flux, into electric signals through the interface circuitry.
Interface circuitry	These signals are conditioned to the appropriate level before being fed into the processor.
Processor	The processor output signals are usually amplified via the interface circuitry to drive power semiconductor devices of the power converter.

The choice of electric propulsion systems for EVs and HEVs mainly depends on a number of factors:

The driver's expectation	The driver's expectation is defined by a driving profile, which includes the acceleration, maximum speed, climbing capability, braking, and range
Vehicle Constraints	The vehicle constraints, including volume and weight, depend on the vehicle type, vehicle weight, and payload.
Energy Source	The energy source relates to batteries, fuel cells, ultracapacitors, flywheels, and various hybrid sources.

Thus, the process of identifying the preferred feature and package options for electric propulsion has to be carried out at the system level. The interaction of subsystems and the likely impacts of system trade-offs must be examined

Electric Vehicle (EV) Drivetrain Alternatives Based on Drivetrain Configuration

There are many possible EV configurations due to the variations in electric propulsion and energy sources. Based on these variations, six alternatives are possible as shown in **Figure 3**. These six alternatives are

- In **Figure 3a** a single EM configuration with gearbox (GB) and a clutch is shown. It consists of an EM, a clutch (C), a gearbox, and a differential (D). The clutch enables the connection or disconnection of power flow from EM to the wheels. The gear consists of a set of gears with different gear ratios. With the use of clutch and gearbox, the driver can shift the gear ratios and hence the torque going to the wheels can be changed. The wheels have high torque low speed in the lower gears and high-speed low torque in the higher gears.
- In **Figure 3b** a single EM configuration without the gearbox and the clutch is shown. The advantage of this configuration is that the weight of the transmission is reduced. However, this configuration demands a more complex control of the EM to provide the necessary torque to the wheels.
- **Figure 3c** shows a configuration of EV using one EM. It is a transverse front EM front wheel drive configuration. It has a fixed gearing and differential and they are integrated into a single assembly.
- In **Figure 3d** a dual motor configuration is shown. In this configuration the differential action of an EV when cornering can be electronically provided by two electric motors.
- In order to shorten the mechanical transmission path from the EM to the driving wheel, the EM can be placed inside a wheel. This configuration is called in-wheel drive. **Figure 3e** shows this configuration in which fixed planetary gearing is employed to reduce the motor speed to the desired wheel speed.
- In **Figure 3f** an EV configuration without any mechanical gearing is shown. By fully abandoning any mechanical gearing, the in-wheel drive can be realized by installing a low speed outer-rotor electric motor inside a wheel.

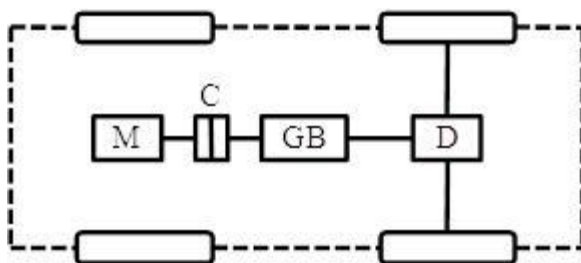


Figure 3a: EV configuration with clutch, gearbox and differential

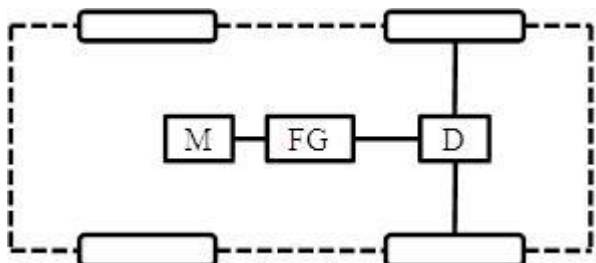


Figure 3b: EV configuration without clutch and gearbox

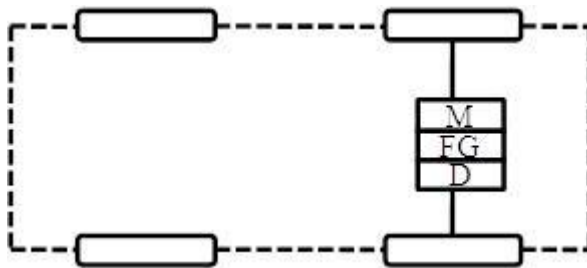


Figure 3c: EV configuration with clutch, gearbox and differential

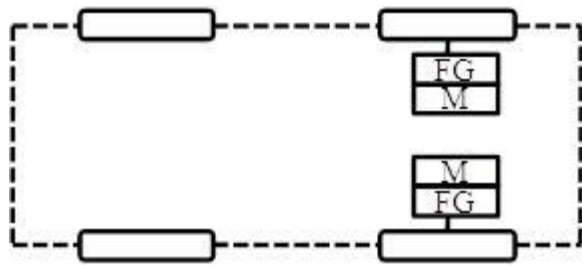


Figure 3d: EV configuration with two EM

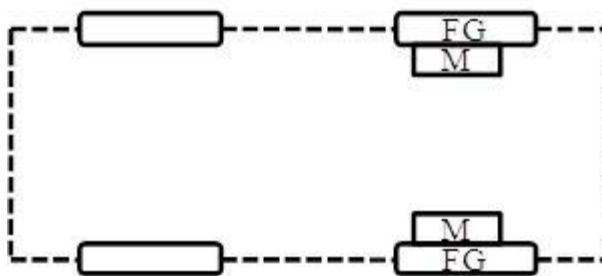


Figure 3e: EV configuration with in wheel motor and mechanical gear

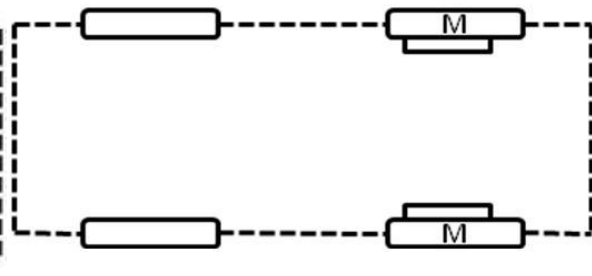


Figure 3f: EV configuration with in wheel motor and no mechanical gear

C: Clutch

D: Differential

FG: Fixed gearing

GB: Gearbox

EM: Electric motor

Electric Vehicle (EV) Drivetrain Alternatives Based on Power Source Configuration

Besides the variations in electric propulsion, there are other EV configurations due to variations in energy sources. There are five configurations possible and they are:

- **Configuration 1:** It is a simple battery powered configuration, **Figure 4a**. The battery may be distributed around the vehicle, packed together at the vehicle back or located beneath the vehicle chassis. The battery in this case should have reasonable specific energy and specific power and should be able to accept regenerative energy during braking. In case of EVs, the battery should have both high specific energy and specific power because high specific power governs the driving range while the high power density governs the acceleration rate and hill climbing capability.
- **Configuration 2:** Instead of two batteries, this design uses two different batteries, **Figure 4b**. One battery is optimized for high specific energy and the other for high specific power.
- **Configuration 3:** In this arrangement fuel cell is used, **Figure 4c**. The battery is an energy storage device, whereas the fuel cell is an energy generation device. The operation principle of fuel cells is a reverse process of electrolysis. In reverse and electrolysis, hydrogen and oxygen gases combine to form electricity and water. The hydrogen gas used by the fuel cell can be stored in an on-board tank whereas oxygen gas is extracted from air. Since fuel cell can offer high specific energy but cannot accept regenerative energy, it is preferable to combine it with battery with high specific power and high-energy receptivity.
- **Configuration 4:** Rather than storing it as a compressed gas, a liquid or a metal hydride, hydrogen can be generated on-board using liquid fuels such as methanol, **Figure 4d**. In this case a mini reformer is installed in the EV to produce necessary hydrogen gas for the fuel cell.
- **Configuration 5:** In fuel cell and battery combination, the battery is selected to provide high specific power and high-energy receptivity. In this configuration a battery and supercapacitor combination is used as an energy source, **Figure 4e**. The battery used in this configuration is a high energy density device whereas the supercapacitor provides high specific power and energy receptivity. Usually, the supercapacitors are of relatively low voltage levels, an additional dc-dc power converter is needed to interface between the battery and capacitor terminals.

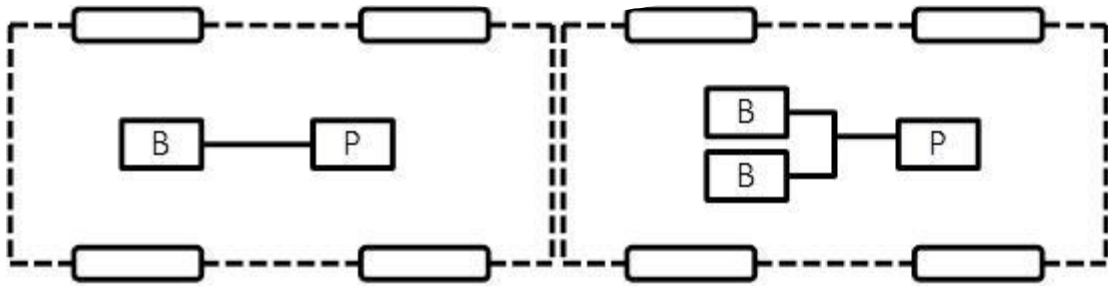


Figure 4a: EV configuration with battery

Figure 4b: EV configuration with two battery sources

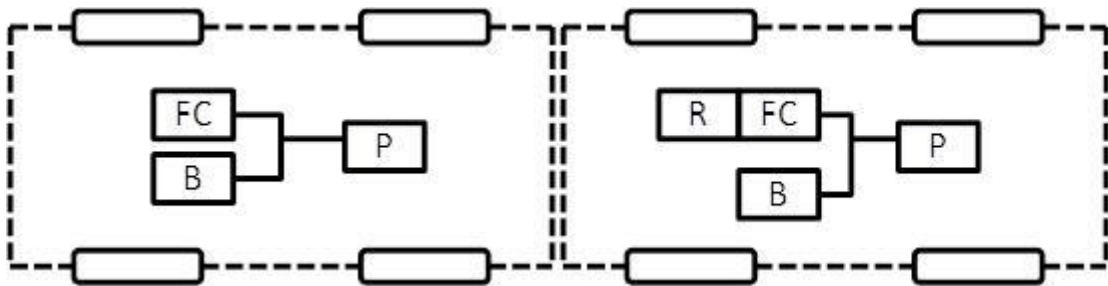


Figure 4c: EV configuration with battery and fuel cell sources

Figure 4d: EV configuration with multiple energy sources

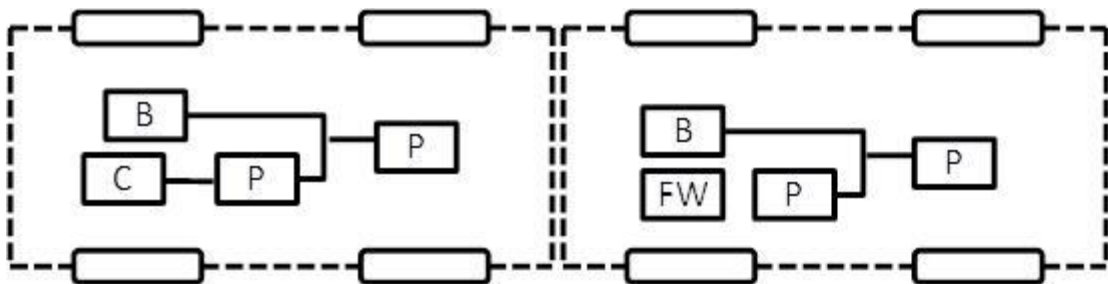


Figure 4e: EV configuration with battery and capacitors sources

Figure 4f: EV configuration with battery and flywheel sources

B: Battery

C: Capacitor

FC: Fuel cell

FW: Flywheel

P: Power converter

R: Reformer

Single and Multi-motor Drives

A differential is a standard component for conventional vehicles. When a vehicle is rounding a curved road, the outer wheel needs to travel on a larger radius than the inner wheel. Thus, the differential adjusts the relative speeds of the wheels. If relative speeds of the wheels are not adjusted, then the wheels will slip and result in tire wear, steering difficulties and poor road holding. In case of EVs, it is possible to dispense the mechanical differential by using two or even four EMs. With the use of multiple EMs, each wheel can be coupled to an EM and this will enable independent control of speed of each wheel in such a way that the differential action can be electronically achieved. In **Figure 5**, a typical dual motor drive with an electronic differential is shown.

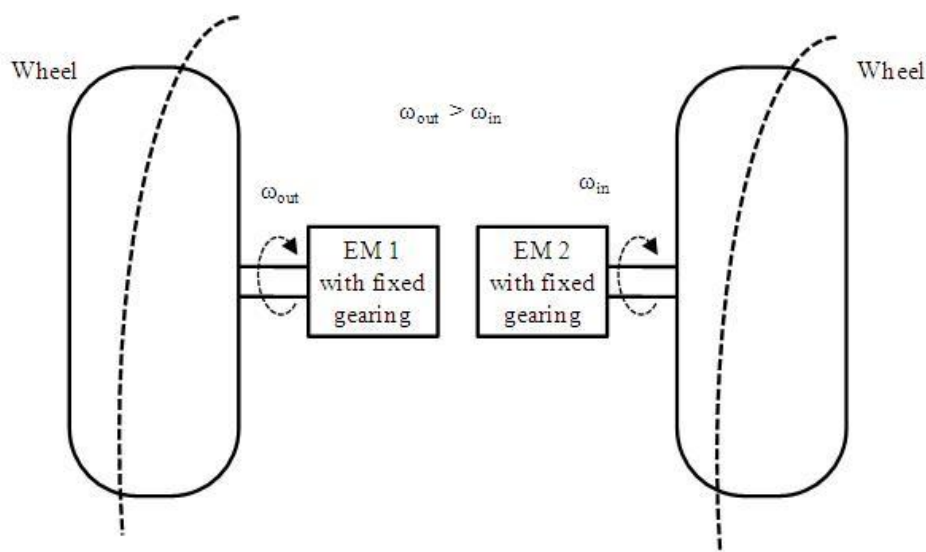


Figure 5: Differential Action

In Wheel Drives

By placing an electric motor inside the wheel, the in wheel motor has the advantage that the mechanical transmission path between the electric motor and the wheel can be minimized. Two possible configurations for in wheel drives are:

- When a high-speed inner-rotor motor is used (**Figure 6a**) then a fixed speed-reduction gear becomes necessary to attain a realistic wheel speed. In general, speed reduction is achieved using a planetary gear set. This planetary gear is mounted between the motor shaft and the wheel rim. Usually this motor is designed to operate up to 1000 rpm so as to give high power density.

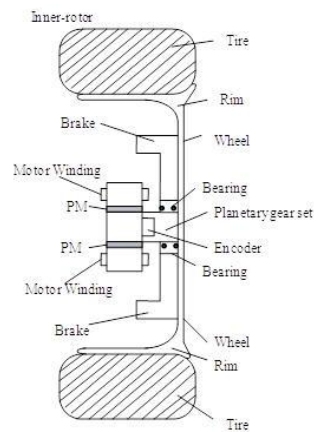


Figure 6a: Inner rotor *In Wheel* drive

- In case outer rotor motor is used (**Figure 6b**), then the transmission can be totally removed and the outer rotor acts as the wheel rim and the motor speed is equivalent to the wheel speed and no gears are required.

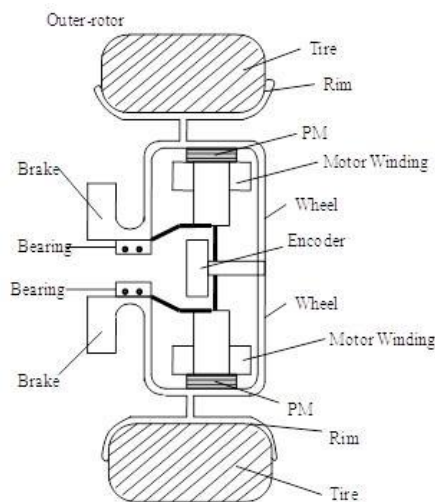


Figure 6b: Outer rotor *In Wheel* drive

The tradeoffs of the high-speed inner rotor motor are:

- It has the advantage of smaller size, lighter weight and lower cost
- Needs additional planetary gearset

The tradeoffs of outer-rotor motor are

- Low speed and hence does not need additional gears
- The drawbacks are larger size, weight and cost because of the low speed design.

Considerations of EMs used in EVs

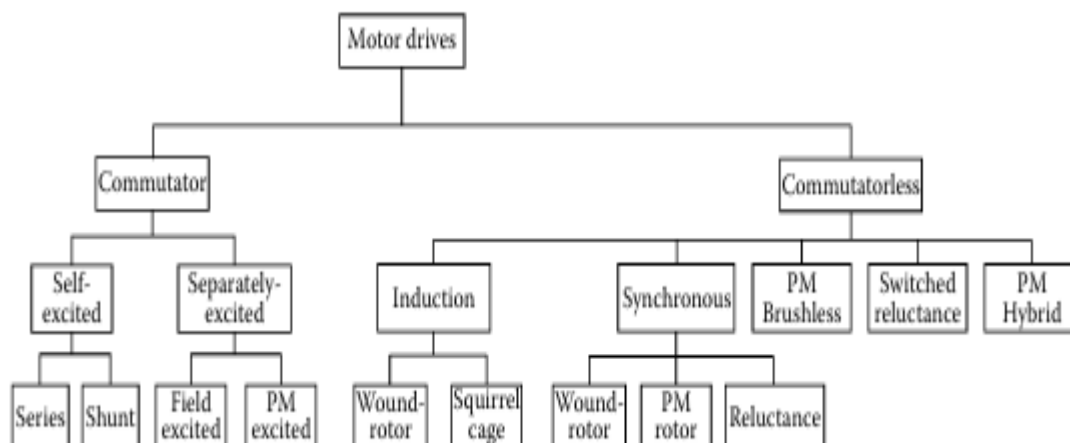
The requirements of EMs used in EVs are:

- Frequent start/stop
- High rate of acceleration and deceleration
- High torque low speed hill climbing
- Low torque cruising
- Very wide speed range of operation

The EMs for EVs are unique and their major differences with respect to industrial motors in load requirement, performance specification and operating environment are as follows:

- EV motors need to produce the maximum torque that is four to five times of the rated torque for acceleration and hill climbing, while industrial motors generally offer the maximum torque that is twice of the rated torque for overload operation
- EV motors need to achieve four to five times the base speed for highway cruising, while industrial motors generally achieve up to twice the base speed for constant power operation
- EV motors require high power density as well as good efficiency map (high efficiency over wide speed and torque ranges), while industrial motors are generally optimized to give high efficiency at a rated point.
- EV motors need to be installed in mobile vehicles with harsh operating conditions such as high temperature, bad weather and frequent vibration, while industrial motors are generally located in fixed places.

The motor drives for EVs and HEVs can be classified into two main groups, namely the commutator motors and commutatorless motors



Commutator motors mainly are the traditional DC motors, which include series excited, shunt excited, compound excited, separately excited, and permanent magnets (PMs) excited motors. DC motors need commutators and brushes to feed current into the armature, thus making them less reliable and unsuitable for maintenance-free operation and high speed. In addition, winding-excited DC motors have low specific power density. Nevertheless, because of their mature technology and simple control, DC motor drives have been prominent in electric propulsion systems.

Technological developments have recently pushed **commutatorless electric motors** into a new era. Advantages include higher efficiency, higher power density, and lower operating cost. They are also more reliable and maintenance-free compared to commutator DC motors; thus, commutatorless electric motors have now become more attractive.

Induction motors are widely accepted as a commutatorless motor type for EV and HEV propulsion. This is because of their low cost, high reliability, and maintenance-free operation. However, conventional control of induction motors such as variable-voltage variable-frequency cannot provide the desired performance. With the advent of the power electronics and microcomputer era, the principle of field-oriented control (FOC) or vector control of induction motors has been accepted to overcome their control complexity due to their nonlinearity. However, these EV and HEV motors using FOC still suffer from low efficiency at light loads and limited constant-power operating range.

By replacing the field winding of conventional **synchronous motors with PMs**, PM synchronous motors can eliminate conventional brushes, sliprings, and field copper losses. Actually, these PM synchronous motors are also called PM brushless AC motors, or sinusoidal-fed PM brushless motors, because of their sinusoidal AC current and brushless configuration. Since these motors are essentially synchronous motors, they can run from a sinusoidal or pulsed waveform modulation supply (PWM supply) without electronic commutation. When PMs are mounted on the rotor surface, they behave as non-salient synchronous motors because the permeability of PMs is similar to that of air. By burying those PMs inside the magnetic circuit of the rotor, the saliency causes an additional reluctance torque, which leads to facilitating a wider speed range at constant power operation. On the other hand, by abandoning the field winding or PMs while purposely making use of the rotor saliency, synchronous reluctance motors are generated. These motors are generally simple and inexpensive, but with relatively low output power. Similar to induction motors, these PM synchronous motors usually use FOC for high-performance applications. Because of their inherently high power density and high efficiency, they have been accepted as having great potential to compete with induction motors for EV and HEV applications.

By virtually inverting the stator and rotor of PM DC motors (commutator), **PM brushless DC (BLDC) motors** are generated. It should be noted that the term “DC” may be misleading, since it does not refer to a DC current motor. Actually, these motors are fed by rectangular AC current and hence are also rectangular-fed PM brushless motors. The most obvious advantage of these motors is the removal of brushes. Another advantage is the ability to produce a large torque

because of the rectangular interaction between current and flux. Moreover, the brushless configuration allows more cross-sectional area for the armature windings. Since the conduction of heat through the frame is improved, an increase in electric loading causes higher power density. Different from PM synchronous motors, these PM BLDC motors generally operate with shaft position sensors. Recently, sensor-less control technologies have been developed in the Power Electronics and Motor Drive Laboratory at Texas A&M University.

Switched reluctance motors (SRMs) have been recognized to have considerable potential for EV and HEV applications. Basically, they are direct derivatives of single-stack variable-reluctance stepping motors. SRMs have the definite advantages of simple construction, low manufacturing cost, and outstanding torque–speed characteristics for EV and HEV applications. Although they possess simplicity in construction, this does not imply any simplicity of their design and control. Because of the heavy saturation of pole tips and the fringe effect of pole and slots, their design and control are difficult and subtle. Traditionally, SRMs operate with shaft sensors to detect the relative position of the rotor to the stator. These sensors are usually vulnerable to mechanical shock and sensitive to temperature and dust. Therefore, the presence of the position sensor reduces the reliability of SRMs and constrains some applications. Recently, sensor-less technologies have been developed in the Power Electronics and Motor Drive Laboratory—again, at Texas A&M University. These technologies can ensure smooth operation from zero speed to maximum speed.